Chain length(or) no. of carbon atoms	Root word	Chain length(or) no. of carbon atom	Root word
C_{12}	Dodec-	C_{60}	Hexacont-
C_{13}	Tridec -	C_{70}	Heptacont-
C_{14}	Tetradec-	C_{80}	Octacont-
C_{15}	Pentadec-	C_{90}	Nonacont-
C_{16}	Hexadec-	C_{100}	Hect-

Suffix: There are two types of suffix. They are primary suffix and secondary suffix

Primary suffix: It denotes the saturation/ unsaturation of organic compounds. It is added immediately after the root word. Primary suffix for various saturated and unsaturated carbon chains are as follows:

Table 11.3 Primary suffix for various saturated and unsaturated carbon chains

Name & type of carbon chain	Primary suffix
Saturated, C-C	ane
Unsaturated carbon chain	
one C=C bond	ene
Two C=C bonds	diene
Three C=C bonds	triene
One C≡C bond	yne
Two C≡C bonds	diyne

Secondary suffix: It is used to denote the nature of functional group present in the organic compound. It is added to the primary suffix by removig its terminal 'e'. Secondary suffix names for some functional groups is listed below in table 11.4

Table 11.4 Secondary suffix and prefixes for some functional groups:

Class of organic compounds	Functional group	Prefix	Secondary suffix
Alcohols	—OH	hydroxy-	-ol
Thioalcohols	—SH	mercapto	-thiol
Aldehydes	—C(=O)—H	formyl-	-al
Ketones	>C=O	oxo-	-one
Carboxylic acid	—C(=O)—OH	carboxy-	-oic acid
Esters	—C(=O)—OR	Alkoxy–Carbonyl	-oate
Acid chlorids	—C(=O)—Cl	chloro-carbonyl	-oyl chloride
Acid amides	—C(=O)—NH_2	Carbam oyl	-amide
Amines	—NH_2	amino-	-amine
Nitriles	—C≡N	cyano-	-nitrile
Sulphonic acid	—SO_3H	sulpho-	-sulphonic acid

Unit 11 revised - jagan.indd 115 22-01-2021 12:42:17